

AI-Assisted Triage in the Emergency Department

Paper 1: Development and Validation of a Practical Machine-Learning Triage Algorithm (2021)

Aim

To determine whether a machine-learning model could improve the identification of Emergency Department patients requiring critical care.

Method

- Retrospective analysis of Emergency Department data.
- Machine-learning model developed and tested.
- Short prospective clinical evaluation.

Key Findings

- Improved identification of critically ill patients.
- Reduced under-triage compared with standard methods.
- Explainable outputs increased clinician confidence and engagement.

Limitations

- Single-centre study.
- Limited prospective testing.
- Some important clinical variables were unavailable.

Significance

Demonstrated that AI can improve triage decisions and patient safety in Emergency Department settings.

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Paper 2: External Validation of the Score for Emergency Risk Prediction (SERP) (2022)

Aim

To assess whether an AI-based triage score remained accurate when applied outside the hospital where it was originally developed.

Method

- External validation study using data from a different healthcare setting.
- Evaluated predictive performance and interpretability.

Key Findings

- Maintained strong predictive accuracy.
- Demonstrated that interpretable AI can support triage decisions.
- Provided evidence that the model could generalise beyond its development site.

Limitations

- Retrospective study design.
- Limited geographical diversity.
- No assessment of real-world implementation.

Significance

Highlighted the importance of validating AI systems before wider clinical adoption.

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Paper 3: Inter-Hospital External Validation of an Interpretable Machine-Learning Triage Score (2024)

Aim

To evaluate whether an AI triage model could perform consistently across multiple hospitals.

Method

- Multi-centre retrospective study.
- Used a Common Data Model to standardise data across hospitals.
- Predicted short-term mortality risk.

Key Findings

- Strong performance across different hospitals.
- Demonstrated good transportability and robustness.
- Showed the value of standardised data for AI validation.

Limitations

- Conducted within one country.
- Retrospective validation only.
- Did not assess patient outcomes after implementation.

Significance

Provided stronger evidence that AI triage tools can generalise across organisations, supporting future large-scale validation studies.

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Problem Statement

Emergency Departments face rising demand, overcrowding and increasing pressure to identify seriously unwell patients quickly and accurately. Artificial intelligence (AI) offers a way to support triage decisions and improve the early recognition of patients at risk of deterioration. Research shows that AI can improve triage accuracy and reduce under-triage. However, hospitals still lack clear evidence that these systems work consistently across different settings and patient groups. Healthcare leaders must decide whether current AI triage tools are safe, reliable and effective enough to support routine clinical practice while maintaining patient safety and clinical accountability.

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Research Gap and Early Direction

Current evidence shows that AI can improve triage accuracy and identify patients at risk of deterioration. However, most studies rely on retrospective data and focus on model performance rather than real-world clinical outcomes. Researchers have not yet demonstrated whether AI-assisted triage improves patient outcomes, reduces waiting times, or supports safer decision-making across diverse healthcare settings. Future research should focus on prospective, multi-centre studies that evaluate the impact of AI on patient safety, clinical workflow, staff experience and service performance in routine Emergency Department practice.

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References

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